THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 13: **Statistics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 3, 16, 20, 4, 7 respectively, The mean of the data is: 1

(a) 44.2 **(b)** 45 **(c)** 45.2 **(d)** 442

2. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 with frequencies 4, 3, 2, 4, 8, 4 respectively, The median of the data is: 1

(a) 70 **(b)** 80 **(c)** 73.4 **(d)** 78.75

3. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90 with frequencies 9, 4, 10, 13, 4 respectively, The mode of the data is: 1

(a) 64.75 **(b)** 70 **(c)** 80 **(d)** 72.5

4. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 3, 8, 20, 12, 7 respectively, The modal class is: 1

(a) 20–30 **(b)** 30–40 **(c)** 40–50 **(d)** 10–20

5. The mean of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 3, p, 20, 4, 7 respectively is 44.2. The value of p is: 1

(a) 21 **(b)** 15 **(c)** 16 **(d)** 18

6. The class mark of the class 10–25 is: 1

(a) 35 **(b)** 7.5 **(c)** 17.5 **(d)** 15

7. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50 with frequencies 8, 16, 8, 10, 8 respectively, The mean of the data is: 1

(a) 25 **(b)** 238 **(c)** 24.8 **(d)** 23.8

8. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90 with frequencies 2, 17, 12, 10, 9 respectively, The median of the data is: 1

(a) 70 **(b)** 65 **(c)** 66.4 **(d)** 60

9. For the data with classes 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 3, 13, 5, 16, 7, 6 respectively, The mode of the data is: 1

(a) 80 **(b)** 80.8 **(c)** 90 **(d)** 85.5

10. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 5, 4, 10, 4, 2 respectively, The modal class is: 1

(a) 20–30 **(b)** 30–40 **(c)** 10–20 **(d)** 40–50

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The marks obtained by students in a test are given below. 2

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	20	22	20	17	6	15

Find the mean.

12. The weights (in kg) of students of a class are given below. 2

Weight (kg)	70–80	80–90	90–100	100–110	110–120
Number of students	13	16	7	7	7

Find the mode.

13. The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. 2

Lifetime (hours)	0–10	10–20	20–30	30–40	40–50	50–60
Number of bulbs	16	22	20	5	19	18

Find the median.

14. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	60–70	70–80	80–90	90–100	100–110
Number of workers	12	20	3	2	3

Find the mean.

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The marks obtained by students in a test are given below. 3

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	20	22	20	17	6	15

Find the mean.

16. The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	0–10	10–20	20–30	30–40	40–50
Number of patients	13	15	9	8	5

Find the median.

17. The daily electricity consumption (in units) of households in a colony is given below. 3

Units consumed	20–30	30–40	40–50	50–60	60–70
Number of households	3	16	20	4	7

Find the mode.

OR

The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	60–70	70–80	80–90	90–100	100–110
Number of workers	12	20	3	2	3

Find the mean.

18. The weights (in kg) of students of a class are given below. 3

Weight (kg)	10–20	20–30	30–40	40–50	50–60
Number of students	2	4	2	15	2

Find the median.

OR

The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. 3

Lifetime (hours)	0–10	10–20	20–30	30–40	40–50
Number of bulbs	8	16	8	10	8

Find the mode.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. The mean of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 and frequencies 4, 3, 2, p, 8, 4 respectively is 73.4. Find the value of p. 5

OR

The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 15, 3, x, 16, y, 10, 13 respectively, and total frequency 100, is 36.25. Find the value of x. 5

20. The mean of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 2, 17, p, 10, 9 respectively is 66.4. Find the value of p. 5

OR

The median of a distribution with classes 30–40, 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 and frequencies 15, 10, 19, x, 16, y, 16 respectively, and total frequency 100, is 71.25. Find the value of x. 5

END OF QUESTION PAPER Total: 40 Marks